EXCERCISES DISCUSSED IN CLASS

1. Infinite coin tosses

The probability space for infinite coin tosses is $(\Omega, \mathcal{F}, \mathbb{P})$. Here

$$\Omega = \{\underline{x} = (x_1, x_2, \ldots) : \text{each } x_i \in \{-1, 1\}\}.$$

$\mathcal{F}$ is the σ-algebra generated by all *cylinder sets.* For $n \in \mathbb{N}$ and $a \in \{-1, 1\}^n$, define the cylinder set $\mathcal{C}_{a,n}$ by:

$$\mathcal{C}_{a,n} = \{\underline{x} \in \Omega : x_i = a_i, \ \forall 1 \leq i \leq n\}.$$

$\mathcal{F} = \sigma\left(\{\mathcal{C}_{a,n} : a \in \{-1,1\}^n\}_{n=1}^{\infty}\right)$. We set $\mathbb{P}(\mathcal{C}_{a,n}) = 1/2^n$. Caratheodary's extension theorem guarantees a unique extension of $\mathbb{P}$ to a probability measure on $(\Omega, \mathcal{F})$; by a slight abuse of notation we call the extension also $\mathbb{P}$.

If we let $S_0 = 0$, $S_n = \sum_{k=1}^n x_k$, then $(S_k)_{k=0}^{\infty}$ denotes the simple random walk on $\mathbb{Z}$ starting at the origin 0.

Excercises. Show that the maps $(\Omega, \mathcal{F}) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ defined in the following way are random variables.

(1) For $n \in \mathbb{N}$, $T(\underline{x}) := x_n$.

(2) For $n \in \mathbb{N}$, $T(\underline{x}) := \sum_{k=1}^n x_k = S_n$.

(3) *Hitting time:* For $a \in \mathbb{Z}$, define

$$\tau_a := \inf\{n > 0 : S_n = a\}.$$

(4) *Exit time:* For $a, b \in \mathbb{Z}$ and $a < b$, define

$$E_{[a,b]} := \inf\{n > 0 : S_n > b \text{ or } S_n < a\}.$$

(5) *Occupation time:* For $a, b, c \in \mathbb{Z}$ and $a < c < b$, define

$$O_c := \#\{0 < n < E_{[a,b]} : S_n = c\}.$$

2. 7 Aug Notes

Notation. (Delta measure) For a measurable space $(\Omega, \mathcal{F})$, let $p \in \Omega$ such that $\{p\} \in \mathcal{F}$. The probability measure δ_p is defined by the foll. for every $A \in \mathcal{F}$:

$$\delta_p(A) = \begin{cases} 1, & \text{if } p \in A, \\ 0, & \text{if } p \notin A. \end{cases}$$

Ex. (Convex combination of p.m. is a p.m.) Let μ_i be probability measures on $(\Omega, \mathcal{F})$. Show that $\sum_{i=1}^n a_i \mu_i$ is again a p.m. if $a_i \geq 0$ and $\sum_{i=1}^n a_i = 1$.

Pushforward measure.

Definition. Let $(\Omega, \mathcal{F}, \mathbb{P})$ be a probability space, and let $(\tilde{\Omega}, \tilde{\mathcal{F}})$ be a measurable space. Suppose $T : (\Omega, \mathcal{F}) \to (\tilde{\Omega}, \tilde{\mathcal{F}})$ is a measurable map. Then the **push forward of** $\mathbb{P}$ **by T** (or more simply the **push forward measure**), denoted $\mathbb{P} \circ T^{-1}$, is a probability measure on $(\tilde{\Omega}, \tilde{\mathcal{F}})$ given by

$$\mathbb{P} \circ T^{-1}(A) := \mathbb{P}(T^{-1}(A)),\ \forall A \in \tilde{\mathcal{F}}.$$

The push forward measure $\mathbb{P} \circ T^{-1}$ is also called the **distribution of** T.

Ex. Show that $\mathbb{P} \circ T^{-1}$ defined above is actually a probability measure on $(\tilde{\Omega}, \tilde{\mathcal{F}})$.

Ex. Consider $T : ([0,1], \mathcal{B}[0,1], \lambda) \to \mathbb{R}$ in the following cases, show that they are measurable and find the push forward measures:

(1) $T = 2 \cdot \mathbb{1}_{[0,\frac{1}{4}]} + 5 \cdot \mathbb{1}_{[\frac{1}{2},1]}$.

(2) $T(x) = 1 - x$.

Remark. Suppose $T : (\Omega, \mathcal{F}, \mathbb{P}) \to (\mathbb{R}^n, \mathcal{B}(\mathbb{R}^n))$ is a ($\mathbb{R}^n$ valued) random variable, then the push forward $\mathbb{P} \circ T^{-1}$ *contains all information about* T. What do we mean by this?

Usually there are several ways to model a random experiment. Here is a simple illustration of this for the experiment of tossing a fair coin twice.

- Let $\Omega = \{(x_1, x_2) : x_1, x_2 \in \{0,1\}\}$, $\mathcal{F} = 2^{\Omega}$, and $\mathbb{P}(\{(x_1, x_2)\}) = 1/4$ for every $(x_1, x_2) \in \Omega$. Here we interpret 0 as a tail and 1 as a head.

- Consider $([0,1], \mathcal{B}[0,1], \lambda)$ and define two random variables X_1, X_2:

$$X_1 = \mathbb{1}_{[0,\frac{1}{2}]},\ X_2 = \mathbb{1}_{[0,\frac{1}{4}]} + \mathbb{1}_{[\frac{1}{2},\frac{3}{4}]}.$$

Ex. In the second case above, find the push forward measure $\lambda \circ T^{-1}$, where $T = (X_1, X_2)$.

The above exercise shows that both (x_1, x_2) and (X_1, X_2) model the same experiment! Any possible question (eg., what is the probability that at least one head shows up in the two tosses) you can ask about this experiment can be answered by knowing the distribution of these random variables. The actual probability space on which these were defined play no role at all!

Here is another example (Gambler's ruin): Consider the SRW (starting at 0) we introduced in §1. For $a > 0$ and $b < 0$ integers, and let τ_a and τ_b be the hitting times defined in §1. Suppose you make a bet with your friend: you win if the SRW reaches a before it reaches b, else she wins. Suppose you are interested in the question of who wins, how long it takes to win etc. All this can be answered just by knowing the distribution μ of (τ_a, τ_b).

Ex. Write the probabilities of the following events in terms of μ:

- Your friend wins the bet.

- You win the bet, but this doesn't happen in the first 10 units of time.

Possible pushforward measures.

Ex. Let $\Omega_n = \{1, 2, \ldots, n\}$, $\mathcal{F}_n = 2^{\Omega_n}$, and let $\mathbb{P}_n$ denote the uniform probability measure on $(\Omega_n, \mathcal{F}_n)$. What are the conditions on n, m so that there is a map $T : \Omega_n \to \Omega_m$ which pushes forward $\mathbb{P}_n$ to $\mathbb{P}_m$?

Ex. Describe all possible push forward measures by random variables $T : (\Omega, \mathcal{F}, \mathbb{P}) \to \mathbb{R}$ in the following scenarios:

(1) $(\Omega, \mathcal{F}, \mathbb{P}) = ([0,1],\ \mathcal{B}[0,1],\ 0.1 \cdot \delta_{1/2} + 0.9 \cdot \delta_{3/4})$.

(2) $\Omega = [0,1]$, $\mathcal{F} = \sigma\left([0, \frac{1}{2}], (\frac{1}{2}, 1]\right)$, and $\mathbb{P}$ is any p.m. on $(\Omega, \mathcal{F})$.

Remark. The above exercises hint at the fact that unless both the σ-algebra $\mathcal{F}$ and the p.m. $\mathbb{P}$ are *rich* enough, it is impossible to get an arbitrary p.m. on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ as the push forward of $(\Omega, \mathcal{F}, \mathbb{P})$.

Theorem 1. Let μ be any Borel p.m. on $\mathbb{R}^n$, then there is a measurable function $T : ([0,1], \mathcal{B}[0,1], \lambda) \to \mathbb{R}^n$ such that the push forward of λ by T is μ. That is, $\lambda \circ T^{-1} = \mu$.

Remark. In the next class, we will prove this theorem for the case $n = 1$.

Cumulative distribution function.

Definition. For μ a Borel p.m. on $\mathbb{R}$, its **cumulative distribution function** (CDF) $F_\mu : \mathbb{R} \to [0,1]$ is the function defined by:

$$F_\mu(x) := \mu((-\infty, x)).$$

Ex. Draw the graph of the CDF for the following Borel p.m. on $\mathbb{R}$: $\mu = \delta_1$, $\mu = 0.1\delta_1 + 0.4\delta_2 + 0.5\delta_3$, $\mu = \lambda$ (the Lebesgue measure).

Lemma 2.1 (Properties of a CDF)**.** Let μ be a p.m. on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Then its CDF F_μ satisfies the following properties:

(C1) F_μ is a non-decreasing function.

(C2) $\lim_{x\to\infty} F_\mu(x) = 1$ and $\lim_{x\to-\infty} F_\mu(x) = 0$.

(C3) F_μ is right continuous.

We will prove the following theorem in the next class.

Theorem 2. Given a function $F : \mathbb{R} \to [0,1]$ satisfying conditions (C1)-(C3), there is a unique Borel p.m. on $\mathbb{R}$ whose CDF is F.

The above result shows that every Borel p.m. on $\mathbb{R}$ is completely determined by its CDF. Also, given a function $F : \mathbb{R} \to [0,1]$ satisfying conditions (C1)-(C3), it is necessarily the CDF of a Borel p.m. on $\mathbb{R}$.

Thus there is a nice one-one correspondence between Borel p.m. on $\mathbb{R}$ and functions $F : \mathbb{R} \to [0,1]$ satisfying (C1)-(C3). This correspondence is very useful because in several instances, it is much easier to deal with monotone functions than measures.